A CASE-STUDY BASED REPORT

ON

Climate Pattern Clustering Using K-Means and Hierarchical Clustering

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Abstract

Climate data contains valuable information that helps in understanding weather behaviour and environmental change. This project, *Climate Pattern Clustering Using K-Means and Hierarchical Clustering*, applies unsupervised machine learning to group climate observations that share similar characteristics. The dataset (Indian Climate Dataset 2024–2025, 7,310 daily records for 10 Indian cities) provides temperature, humidity, rainfall, wind speed and atmospheric pressure, among other attributes. After preprocessing (checking data quality, log-transforming the skewed rainfall variable, removing redundant temperature columns and standardizing features), K-Means and Agglomerative Hierarchical Clustering (Ward linkage) are used to discover climate patterns without any predefined labels.

Both algorithms are evaluated with the Silhouette score, Davies–Bouldin index and Calinski–Harabasz index, and are compared visually through PCA projections and a dendrogram. K-Means gave the better and more stable result (Silhouette 0.184 vs 0.136; Davies–Bouldin 2.10 vs 2.34). Both methods consistently separate **rainy** days from **dry** days, and agree on 85.2% of the records. The study also shows, honestly, that this dataset has weak natural cluster structure (Silhouette below 0.20 for every k tested) and that its 10 cities have almost identical climate statistics, so cities themselves are not recovered as climate zones. The project demonstrates how clustering supports climate analysis, environmental monitoring and decision-making for agriculture and disaster management.

1. Introduction

1.1 Background

Climate analysis plays an important role in understanding environmental change and anticipating future weather conditions. With the rapid growth of climate data, traditional manual or purely statistical analysis is often insufficient to expose hidden relationships among environmental factors such as temperature, humidity, rainfall, wind speed and pressure. Machine learning offers efficient techniques for analysing such data at scale.

Clustering is an *unsupervised* learning task: given unlabeled data, the algorithm groups similar observations together so that records inside a group are more alike than records in different groups. Two widely used families are partition-based clustering (K-Means) and hierarchical clustering (agglomerative, with a dendrogram). Earlier studies have used both on meteorological data:

Study (as reviewed in project PPT)	Algorithm	Key finding
Climate classification, India	K-Means	Identified distinct climate types in India; k = 6 gave the highest silhouette score (0.30) and improved climate classification.
Meteorological conditions, Spain	K-Means + Hierarchical	Clustering identified meaningful weather conditions; comparing several algorithms with evaluation measures helps select the best cluster quality.
Rainfall clustering, Mediterranean	K-Means + Hierarchical (Ward)	Both methods produced similar classifications and identified different rainfall-distribution patterns over time.

1.2 Motivation for the Project

Traditional monitoring systems mainly report *current* conditions; they do not reveal which days or regions actually share similar climate behaviour. This creates three practical challenges: (1) difficulty spotting regional climate similarities, (2) slower environmental monitoring and forecasting, and (3) limited support for agriculture and disaster planning. Grouping similar climate conditions automatically can help farmers plan crops, help authorities prepare for heavy-rain or heat events, and give researchers a compact summary of a large dataset.

1.3 Overview of the Work Carried Out

- Collected and inspected the Indian Climate Dataset 2024–2025 (7,310 rows × 13 columns, 10 cities, 731 days each).
- Performed data-quality checks and exploratory data analysis: summary statistics, histograms, box plots and a correlation matrix.
- Prepared features: dropped redundant columns, applied log transformation to rainfall and standardized all features with z-scores.
- Chose the number of clusters using the elbow method and Silhouette, Davies–Bouldin and Calinski–Harabasz scores for k = 2…10.
- Implemented K-Means and Ward-linkage Hierarchical Clustering, visualized them with PCA and a dendrogram, and profiled the resulting clusters.
- Compared both algorithms quantitatively (internal indices, agreement via Adjusted Rand Index) and qualitatively (interpretability).

1.4 Scope and Responsible-Use Statement

The scope of this study is limited to unsupervised grouping of daily climate records using five meteorological variables. The clusters describe *patterns in the supplied data*; they are not forecasts, official climate classifications or hazard warnings, and must not be used on their own for disaster-management or agricultural decisions. The dataset is a public Kaggle dataset containing no personal information. Results should be validated against official sources such as the India Meteorological Department (IMD) before any real-world use.

2. Problem Statement

2.1 Case-Study Scenario

A regional environmental-monitoring team holds two years of daily weather readings for ten major Indian cities. The readings are numerous, unlabeled and multi-dimensional, so analysts cannot manually tell which days behave alike. The team wants a data-driven way to group similar climate conditions and to decide which clustering technique gives clearer, more actionable groups.

2.2 Central Case-Study Question

Can K-Means and Hierarchical Clustering discover meaningful, interpretable climate patterns in the Indian climate data without predefined labels, and which of the two methods does this more effectively?

2.3 Formal Problem Definition

Let $X = \{x_1, ..., x_n\}$ be n = 7,310 observations, each a 5-dimensional real vector x_i of standardized temperature, humidity, log-rainfall, wind speed and pressure. The task is to partition X into k disjoint clusters $C_1, ..., C_k$ that minimize within-cluster dispersion and maximize between-cluster separation, without using target labels. For K-Means the objective is to minimize J = sum over j = 1..k of the sum over all x in C_j of the squared Euclidean distance $d(x, \mu_j)^2$, where μ_j is the centroid of C_j. For Ward hierarchical clustering, the two clusters whose merge gives the smallest increase in total within-cluster variance are merged at each step. The two partitions are then compared using internal validity indices.

2.4 Objectives that Guide the Study

- Group climate observations by shared characteristics without predefined labels.
- Analyze temperature, humidity, rainfall, wind speed and atmospheric pressure jointly.
- Implement K-Means and Hierarchical Clustering on the same dataset.
- Compare both methods and identify the more effective clustering approach.
- Support climate analysis, environmental monitoring and disaster-management decisions.

2.5 Assumptions and Limitations

- Each daily record is treated as an independent observation; temporal order and seasonality are not modelled.
- Euclidean distance on standardized features is assumed to be a suitable measure of climate similarity.
- Only five features are clustered; AQI and cloud cover are kept for profiling. Clusters change if other features are included.
- Internal validity indices (Silhouette, Davies–Bouldin, Calinski–Harabasz) are used because no ground-truth climate-zone labels exist.
- **Data limitation:** the dataset's statistical behaviour (near-uniform distributions, almost zero correlation between unrelated variables, and nearly identical averages for all 10 cities) suggests it may be simulated rather than measured. Conclusions therefore describe this dataset and may not transfer to real IMD observations.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The **Indian Climate Dataset (2024–2025)** was obtained from Kaggle in CSV format. It contains **7,310 records and 13 columns**: one record per city per day for 731 consecutive days (1 Jan 2024 – 31 Dec 2025) across 10 cities (Mumbai, Delhi, Bengaluru, Chennai, Kolkata, Hyderabad, Ahmedabad, Jaipur, Lucknow, Bhopal), i.e. 731 records per city.

#	Feature	Type	Description
1	Date	Date	Observation date
2	City	Categorical	City name (10 cities)
3	State	Categorical	State of the city
4	Temperature_Max (°C)	Numeric	Daily maximum temperature
5	Temperature_Min (°C)	Numeric	Daily minimum temperature
6	Temperature_Avg (°C)	Numeric	Daily average temperature
7	Humidity (%)	Numeric	Relative humidity
8	Rainfall (mm)	Numeric	Daily rainfall
9	Wind_Speed (km/h)	Numeric	Wind speed
10	AQI	Numeric	Air Quality Index
11	AQI_Category	Categorical	Good / Satisfactory / Moderate / Poor / Very Poor
12	Pressure (hPa)	Numeric	Atmospheric pressure
13	Cloud_Cover (%)	Numeric	Cloud cover

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) checks data quality, reveals ranges, distributions, outliers and relationships between variables, and guides preprocessing choices. For clustering it is especially important because distance-based algorithms are sensitive to feature scale, skewness and redundant features.

3.3 Preparation Steps Actually Performed

- **Data-quality check.** No missing values (0) and no duplicate rows (0) were found; data types were correct.
- **Feature selection.** Temperature_Max and Temperature_Min correlate at 0.90 with each other and 0.97–0.98 with Temperature_Avg, so only Temperature_Avg was kept. Clustering uses five features: Temperature_Avg, Humidity, Rainfall, Wind_Speed and Pressure (as stated in the project abstract). AQI, AQI_Category and Cloud_Cover were excluded from clustering; City, State and Date were excluded as identifiers.
- **Skew handling.** Rainfall is strongly right-skewed (skewness 2.55); 59.2% of days have zero rainfall. A log(1 + x) transformation was applied so that a few heavy-rain days do not dominate distances.
- **Outlier treatment.** IQR rule: the only variable with flagged values is Rainfall (1,040 records, 14.2%). These are genuine heavy-rain events and were retained (after the log transform) rather than deleted.
- **Standardization.** All five features were z-score standardized (StandardScaler) so each contributes equally to Euclidean distance.

3.4 EDA Exploration – Implementation

Table 3.1 – Descriptive statistics of numerical features (n = 7,310)

Feature	Mean	Std	Min	25%	50%	75%	Max
Temperature_Max (°C)	34.95	5.78	25	30	34.9	40	45
Temperature_Min (°C)	25.01	6.48	10.1	19.9	25	30	39.8
Temperature_Avg (°C)	29.98	5.97	17.6	25	30	35	42.3
Humidity (%)	62.65	18.68	30	46.4	62.7	78.7	95
Rainfall (mm)	8.23	17.76	0	0	0	6.3	79.9
Wind_Speed (km/h)	13.52	6.56	2	7.9	13.5	19.1	25
AQI	193.76	89.18	40	116	194	270	349
Pressure (hPa)	1007.36	10.11	990	998.7	1007.3	1016.2	1025
Cloud_Cover (%)	52.63	27.33	5	28.92	52.7	76.2	100

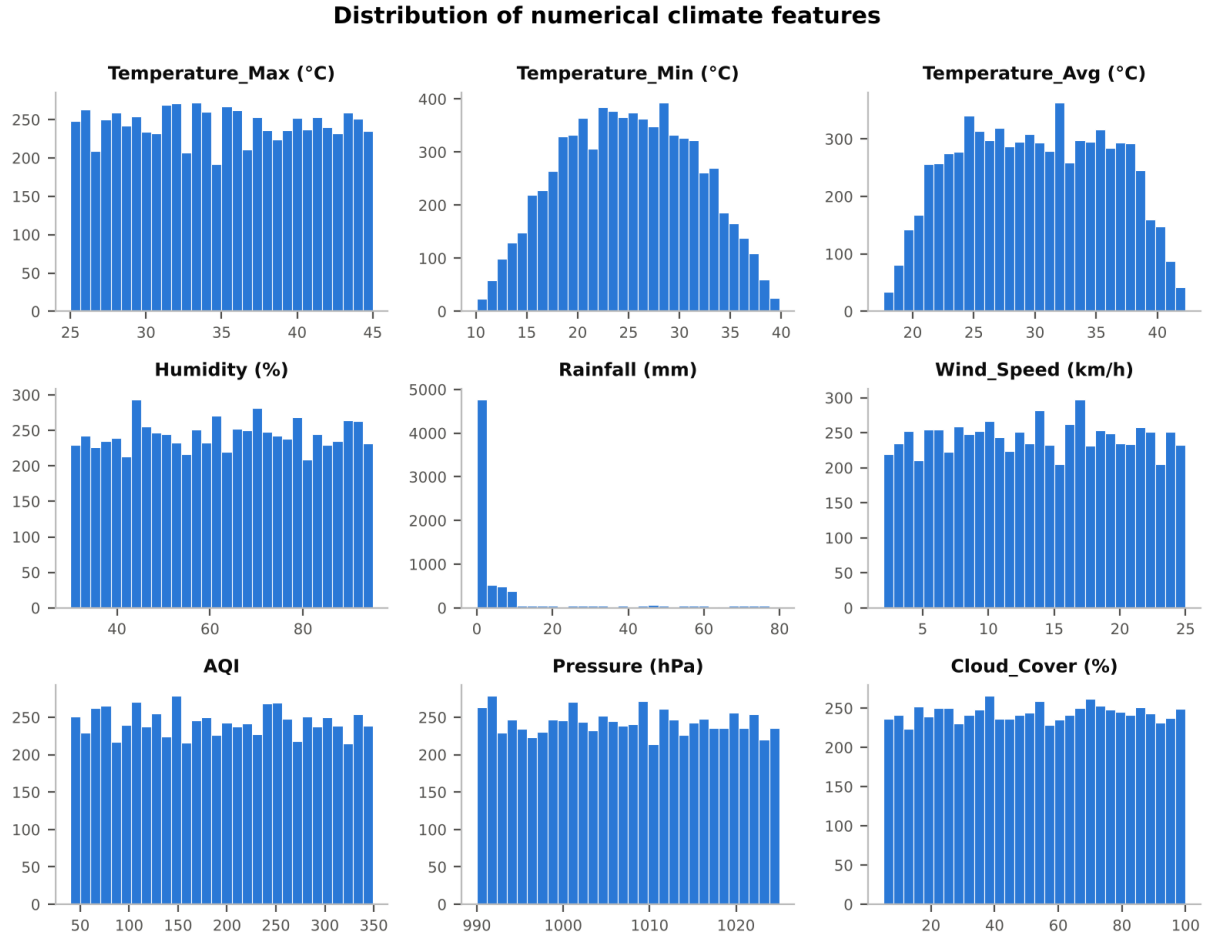


Figure 3.1 – Distribution of numerical climate features. Rainfall is highly right-skewed; other variables are close to uniform.

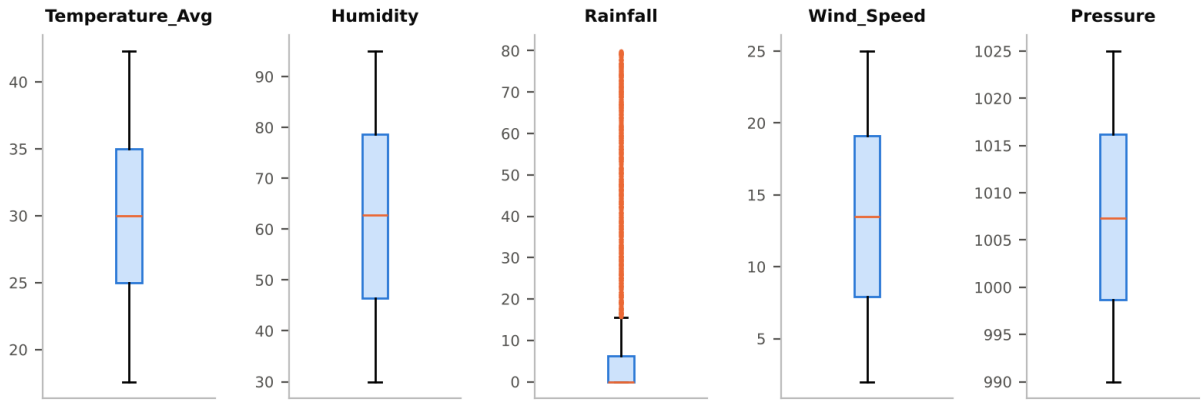


Figure 3.2 – Box plots of the five clustering features. Only rainfall shows points beyond the whiskers.

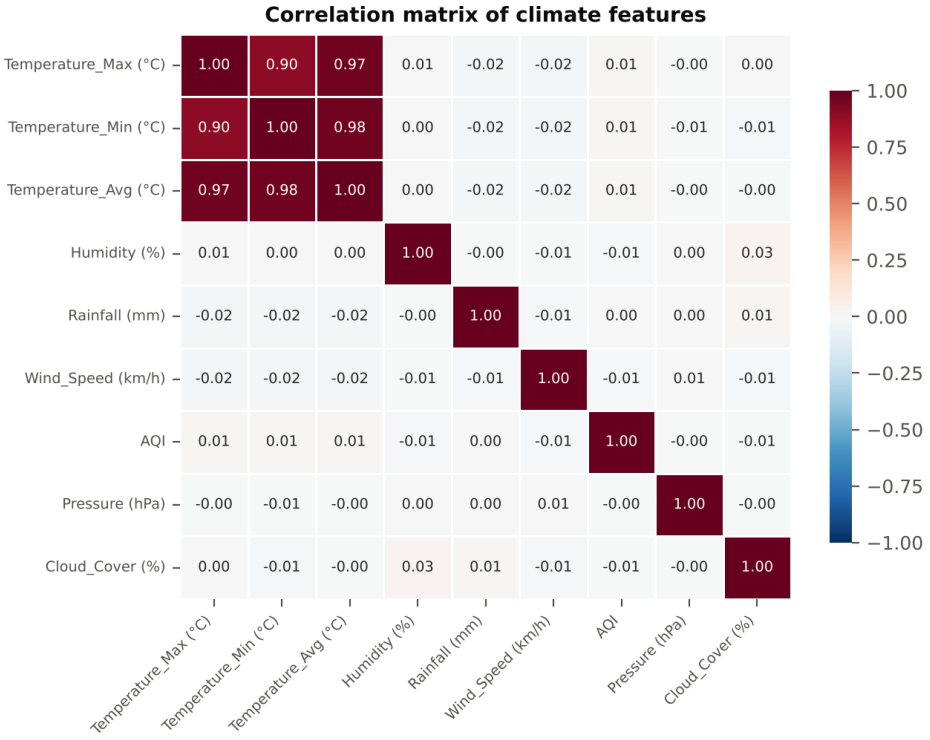


Figure 3.3 – Correlation matrix. Only the three temperature variables are strongly correlated; all other pairs are near zero.

Table 3.2 – Average climate by city

City	Temp avg (°C)	Humidity (%)	Rainfall (mm)	Wind (km/h)	Pressure (hPa)
Ahmedabad	30.1	63.7	8.5	13.5	1007.5
Bengaluru	30.1	62.3	7.5	13.2	1007.0
Bhopal	30.0	63.5	8.3	13.7	1007.6
Chennai	29.8	61.8	7.9	13.5	1006.8
Delhi	30.1	62.6	8.3	13.3	1007.9
Hyderabad	30.0	62.4	8.7	13.3	1007.2
Jaipur	29.6	62.3	8.5	13.7	1007.2
Kolkata	30.4	63.5	8.4	13.5	1007.5
Lucknow	29.9	61.9	8.3	13.7	1007.4
Mumbai	29.8	62.7	8.1	13.8	1007.5

3.5 Summary of EDA Findings

- The dataset is complete and clean: 7,310 observations, 13 attributes, no missing values or duplicates.
- Features have very different scales (pressure ≈ 1007 hPa vs wind ≈ 13.5 km/h), so standardization is mandatory before clustering.
- Rainfall is the only skewed variable (zero on 59% of days, up to 79.9 mm); its extremes look like real weather events, not errors.
- Temperature_Max, Min and Avg are almost redundant ($r \geq 0.90$); keeping all three would triple the weight of temperature in the distance.
- Apart from temperature, features are essentially uncorrelated, and all 10 cities have nearly identical means (temp 29.6–30.4 °C, humidity 61.8–63.7%). Monthly mean temperature varies only between 29.0 and 30.7 °C. This implies weak natural cluster structure and no strong city-wise or seasonal signal.
- Since this is unsupervised learning there is no target variable; clusters are generated by the algorithms and judged with internal validity indices.

4. Algorithm Selection and Implementation

4.1 Why K-Means and Hierarchical Clustering

Both methods need no labels and work on numeric features, but they behave differently, which makes the comparison informative.

Aspect	K-Means	Hierarchical (Agglomerative, Ward)
Idea	Partitions data into k clusters by minimizing distance to cluster centroids.	Builds nested clusters bottom-up by repeatedly merging the closest clusters; shown as a dendrogram.
k required in advance	Yes	No (cut the dendrogram at any level)
Strengths	Fast, scalable, simple centroids that are easy to interpret.	Reveals structure at multiple similarity levels; deterministic.
Weaknesses	Assumes roughly spherical clusters; depends on initialization.	$O(n^2)$ memory/time; merges cannot be undone; harder for large n.

4.2 Software and Hardware Environment

Item	Specification
Operating system	Windows 10/11 or Ubuntu Linux
Language	Python 3.x
Environment	Jupyter Notebook / Google Colab / Visual Studio Code
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn (KMeans, AgglomerativeClustering), SciPy (dendrogram)
Dataset	Indian Climate Dataset 2024–2025 (CSV)
Hardware (minimum)	Intel Core i3 or higher, 4 GB RAM, 500 MB free storage; no GPU required

4.3 Implementation Workflow

Pipeline: **Load CSV → quality checks → feature selection → log-transform rainfall → standardize → choose k → fit K-Means and Ward clustering → evaluate → visualize (PCA, dendrogram) → profile clusters → compare.** Core code (random_state = 42 throughout for reproducibility):

```
import numpy as np, pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score, davies_bouldin_score, calinski_harabasz_score
from scipy.cluster.hierarchy import linkage, dendrogram

df = pd.read_csv("Indian_Climate_Dataset_2024_2025.csv")
feat = ["Temperature_Avg (°C)", "Humidity (%)", "Rainfall (mm)",
        "Wind_Speed (km/h)", "Pressure (hPa)"]
X = df[feat].copy()
X["Rainfall (mm)"] = np.log1p(X["Rainfall (mm)"])      # reduce skew
Z = StandardScaler().fit_transform(X)                  # z-scores

for k in range(2, 11):                                 # choose k
    km = KMeans(k, n_init=10, random_state=42).fit(Z)
    hc = AgglomerativeClustering(k, linkage="ward").fit_predict(Z)
    print(k, km.inertia_, silhouette_score(Z, km.labels_),
          davies_bouldin_score(Z, km.labels_), silhouette_score(Z, hc))

kmeans = KMeans(2, n_init=10, random_state=42).fit(Z)
hier   = AgglomerativeClustering(2, linkage="ward").fit(Z)
dendrogram(linkage(Z, "ward"), truncate_mode="lastp", p=30)
```

4.4 Choosing the Number of Clusters

The number of clusters k was examined from 2 to 10 using the elbow method on K-Means inertia and three internal indices. *Silhouette* (higher is better, range −1 to 1) measures how well a point fits its own cluster against the nearest other cluster; *Davies–Bouldin* (lower is better) measures average cluster overlap; *Calinski–Harabasz* (higher is better) is the ratio of between- to within-cluster dispersion.

Table 4.1 – Cluster-validity indices for k = 2…10 (KM = K-Means, HC = Hierarchical Ward; best value per column in bold)

k	Inertia	KM Sil	KM DB	KM CH	HC Sil	HC DB	HC CH
2	30,592	0.184	2.095	1423	0.136	2.342	874
3	26,558	0.163	1.894	1375	0.102	2.422	879
4	23,885	0.155	1.693	1291	0.104	2.029	892
5	21,648	0.164	1.555	1257	0.099	1.824	882
6	19,906	0.167	1.421	1221	0.092	1.723	858
7	18,644	0.164	1.426	1169	0.095	1.767	839
8	17,522	0.169	1.344	1133	0.106	1.802	820
9	16,595	0.169	1.452	1097	0.095	1.722	799
10	15,580	0.179	1.364	1092	0.093	1.679	779

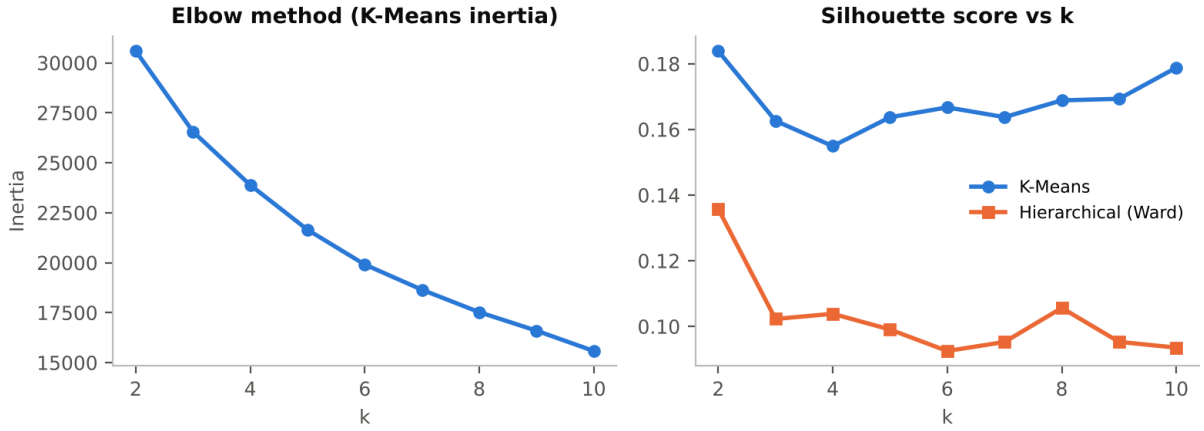


Figure 4.1 – Elbow curve (left) and Silhouette score vs k (right).

The elbow curve decreases smoothly with **no sharp elbow**, which is itself a sign of weak cluster structure. K-Means Silhouette is highest at **k = 2 (0.184)**; Davies–Bouldin and inertia keep improving as k grows, which is expected because more clusters always make groups tighter. Calinski–Harabasz is also highest at k = 2. Since k = 2 is best on both Silhouette and Calinski–Harabasz and gives an easily interpretable split, **k = 2** was used for the final models.

4.5 Results: K-Means vs Hierarchical Clustering

Table 4.2 – Final comparison at k = 2

Metric	K-Means	Hierarchical (Ward)	Better
Silhouette (↑)	0.184	0.136	K-Means
Davies–Bouldin (↓)	2.095	2.342	K-Means
Calinski–Harabasz (↑)	1,423	874	K-Means
Cluster sizes	2418 / 4892	1697 / 5613	–
Agreement between methods	Adjusted Rand Index = 0.48; 85.2% of records get the same group		

As an extra check, other linkages were tried for the hierarchical method at k = 2: Ward (Silhouette 0.136), complete (0.093) and average (0.171). Ward was kept because it is the standard variance-minimizing choice and matches the literature reviewed; average linkage scored slightly higher on Silhouette than Ward but still below K-Means (0.184).

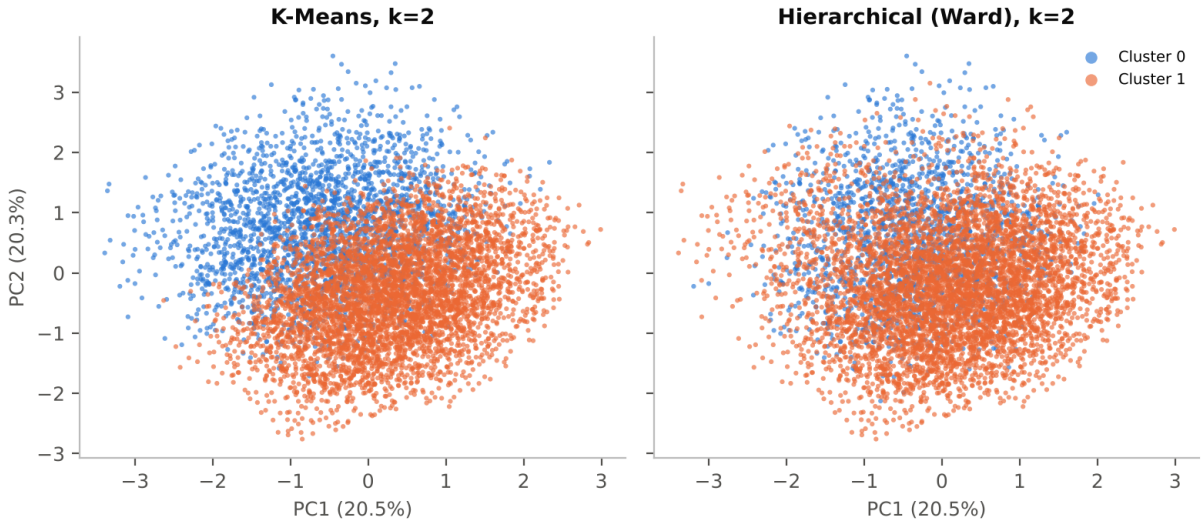


Figure 4.2 – Clusters projected on the first two principal components (PCA explains only 41% of variance, so overlap in 2-D is expected). Colours of the two methods are matched by best overlap.

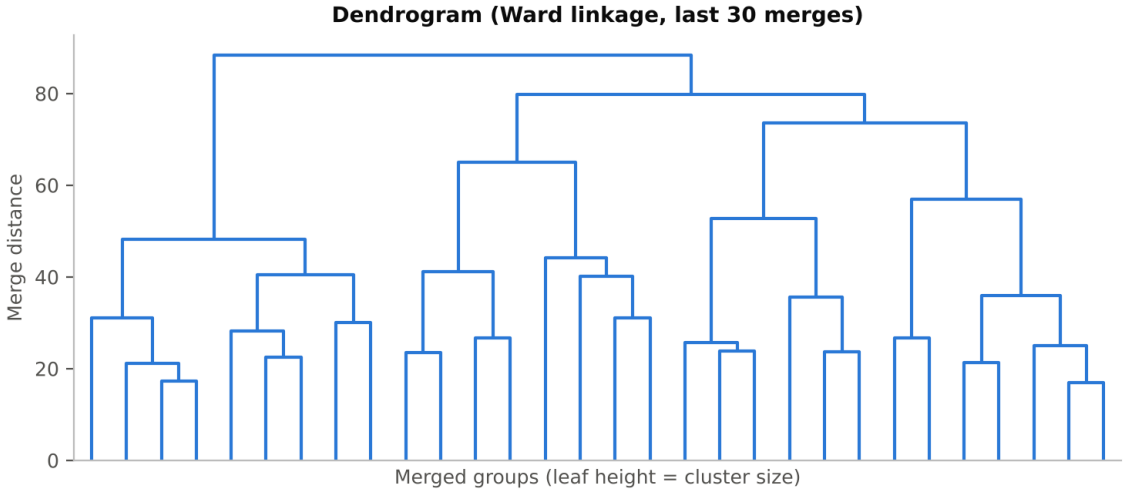


Figure 4.3 – Truncated dendrogram of the Ward linkage; the largest merge distances occur at the top, supporting a small number of clusters.

4.6 Cluster Profiles and Interpretation

Table 4.3 – Mean feature values per cluster (original units)

Cluster	Temp (°C)	Humidity (%)	Rainfall (mm)	Wind (km/h)	Pressure (hPa)	Records
KM-0	30.0	61.8	24.5	13.6	1007.2	2,418
KM-1	30.0	63.1	0.2	13.5	1007.4	4,892
HC-0	29.0	69.8	24.8	12.1	1008.1	1,697
HC-1	30.3	60.5	3.2	14.0	1007.1	5,613

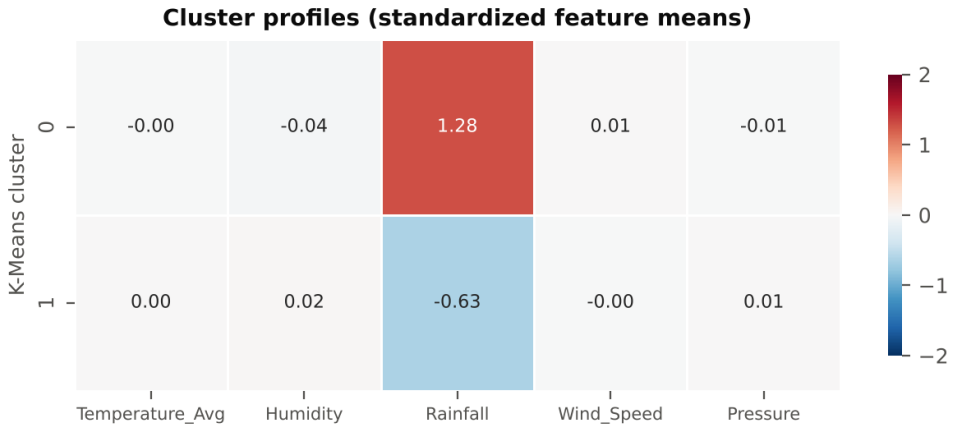


Figure 4.4 – K-Means cluster profiles as standardized feature means (red = above average, blue = below).

- **K-Means cluster 0 – "Rainy days" (2,418 records, 33%).** Average rainfall about 24 mm and slightly lower humidity/wind than average; other variables are near the overall mean.
- **K-Means cluster 1 – "Dry days" (4,892 records, 67%).** Essentially no rain (mean 0.2 mm).

- **Hierarchical clusters** separate a smaller "cooler, humid, rainy" group (1,697 records; 29.0 °C, 70% humidity, 25 mm) from a larger "warmer, drier" group (5,613 records; 30.3 °C, 61% humidity, 3.2 mm). Ward therefore mixes humidity and temperature into the split, whereas K-Means is driven mostly by rainfall.
- **City check.** The Adjusted Rand Index between city and cluster labels is ≈ 0 (-0.000 for K-Means, 0.000 for Hierarchical): each city's days fall roughly 1/3 vs 2/3 in the two clusters (range 31.1–68.9%). The clusters describe *kinds of weather days*, not distinct geographic climate zones.

4.7 Discussion

K-Means outperforms Ward hierarchical clustering on all three internal indices and is also much faster and easier to scale, so it is the more effective method for this dataset. Hierarchical clustering adds value through its dendrogram and by showing how groups merge, and the 85.2% agreement between the two methods shows that the rainy-versus-dry pattern is stable and not an artifact of a single algorithm. However, every Silhouette value is below 0.20, which means the groups overlap considerably: the data forms a continuum rather than sharply separated clusters. This matches the EDA, where features are almost uniformly distributed and uncorrelated. The results should therefore be read as a useful *segmentation* of weather days, not as evidence of naturally distinct climate regimes.

5. Conclusion

- Climate data holds hidden patterns that traditional monitoring alone cannot surface; unsupervised learning can group records by shared climate behaviour with no predefined labels.
- K-Means and Hierarchical Clustering were applied and compared on the same cleaned, log-transformed and standardized dataset of 7,310 daily records.
- K-Means was the better method (Silhouette 0.184, Davies–Bouldin 2.10, Calinski–Harabasz 1,423), while the hierarchical dendrogram gave complementary insight; the two agree on 85.2% of the records.
- Both methods reveal a stable rainy-day vs dry-day structure, which can support agriculture planning, environmental monitoring and disaster-management preparation.
- Cluster quality is modest and the cities are statistically indistinguishable, so the data (which appears simulated) limits what can be concluded; validating the approach on real IMD station data is the key next step.

Future work: use real multi-year observations, add seasonal and spatial features, try DBSCAN or Gaussian Mixture Models, and deploy the cluster labels in a monitoring dashboard.

References

- Indian Climate Dataset 2024–2025, Kaggle (public dataset used in this project).
- Pedregosa et al., "Scikit-learn: Machine Learning in Python", *JMLR* 12, 2011.
- MacQueen, J., "Some methods for classification and analysis of multivariate observations", 1967 (K-Means).
- Ward, J. H., "Hierarchical grouping to optimize an objective function", *JASA* 58, 1963.

- Rousseeuw, P. J., “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis”, *J. Comput. Appl. Math.* 20, 1987.

Signature of the Team members